

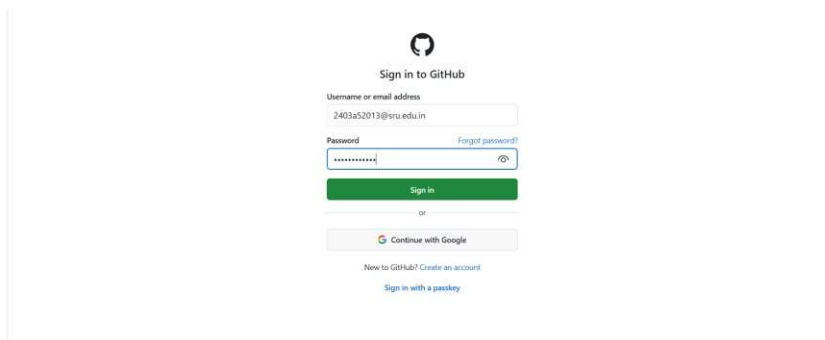
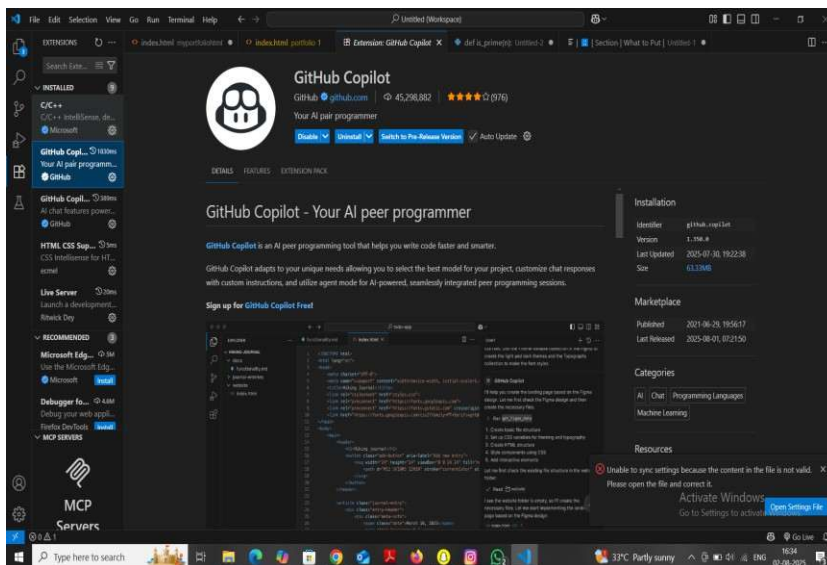
AI Assisted Coding Lab Assignment (1.4/24)

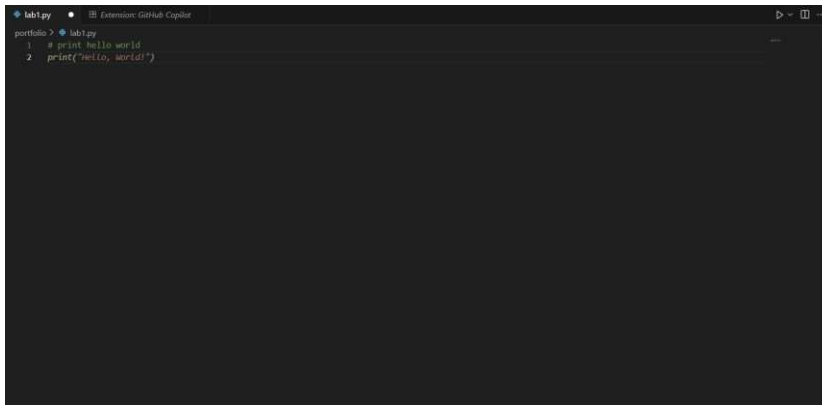
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Section : CS-AI B01

Task 1: GitHub Copilot Setup



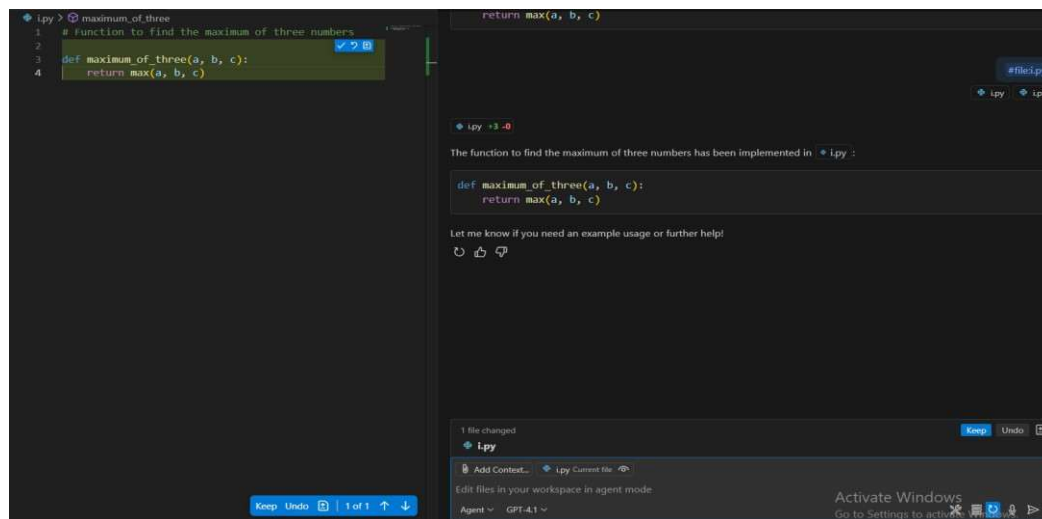


```
lab1.py
portfolio > lab1.py
1 # print hello world
2 print("hello, world!")
```

Task 2: Maximum of Three Numbers

Prompt used:

Function to find the maximum of three numbers Code:



```
maximum_of_three
1 # Function to find the maximum of three numbers
2
3 def maximum_of_three(a, b, c):
4     return max(a, b, c)
```

```
return max(a, b, c)
```

The function to find the maximum of three numbers has been implemented in `maximum_of_three.py` :

```
def maximum_of_three(a, b, c):
    return max(a, b, c)
```

Let me know if you need an example usage or further help!

1 file changed
1.py

Agent: GPT-4.1

Explanation:

The function uses Python's built-in `max()` function to return the largest of the three inputs.

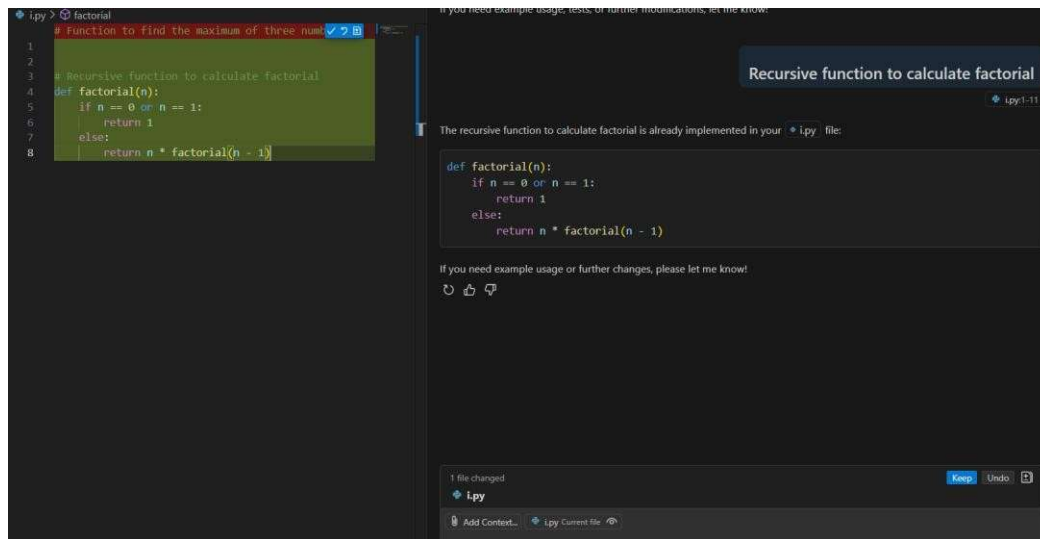
Sample Output:

Maximum is: 25

Task 3: Factorial (Recursive)

Prompt used:

Recursive function to calculate factorial Code:



The screenshot shows a Jupyter Notebook interface. On the left, a code cell contains the following Python code:

```
1 # Function to find the maximum of three num.  
2  
3 # Recursive function to calculate factorial  
4 def factorial(n):  
5     if n == 0 or n == 1:  
6         return 1  
7     else:  
8         return n * factorial(n - 1)
```

On the right, a chat window titled "Recursive function to calculate factorial" shows a response from an AI assistant. The assistant states: "The recursive function to calculate factorial is already implemented in your `lpy` file:" and displays the same code as in the code cell. Below this, the assistant offers further assistance: "If you need example usage or further changes, please let me know!" with icons for refresh, thumbs up, and thumbs down. At the bottom of the chat window, a status bar indicates "1 file changed" and provides options to "Keep", "Undo", or "Add Context...".

Explanation:

The function calls itself recursively until n becomes 1 or 0, which is the base case.

Sample Output:

Factorial of 5 is: 120

Task 4: Student Class

Prompt used:

Define a class Student with name, roll_no, marks and display method

Code:

```
1 ipy > ...
2 |
3 # Define a class Student with name, roll_no, marks a
4 class Student:
5     def __init__(self, name, roll_no, marks):
6         self.name = name
7         self.roll_no = roll_no
8         self.marks = marks
9
10
11     def display(self):
12         print(f"Name: {self.name}")
13         print(f"Roll No: {self.roll_no}")
14         print(f"Marks: {self.marks}")
```

if you need example usage or further changes, please let me know:

Define a class Student with name, roll_no, marks and display method

ipy +13 -0

The `Student` class with `name`, `roll_no`, `marks`, and a `display` method has been added to your `ipy` file.

Let me know if you need an example of how to use this class or any further help!

1 file changed
ipy

Agent GPT-4.1

Explanation:
This class defines a constructor (`__init__`) and a method `display()` to print the details of a student object.

Sample Output:
Name: Aluwala Sharanya
Roll No: 2403A52013
Marks: 89

Task 5: Word Frequency

Prompt used:
Function to return word frequency from a given string

Code:

```
# Function to return word frequency from a given str
def word_frequency(s):
    words = s.split()
    freq = {}
    for word in words:
        freq[word] = freq.get(word, 0) + 1
    return freq
```

Function to return word frequency from a given string

ipy +8 -0

The function to return word frequency from a given string has been added to your `ipy` file.

```
def word_frequency(s):
    words = s.split()
    freq = {}
    for word in words:
        freq[word] = freq.get(word, 0) + 1
    return freq
```

Let me know if you need anything else!

1 file changed
ipy

Agent GPT-4.1

Explanation:

The function splits the input string into words, converts them to lowercase, and uses a dictionary to count how many times each word appears.

Sample Output:

```
{'ai': 2, 'makes': 1, 'coding': 2, 'easier': 1, 'is': 1, 'fun': 1, 'with': 1}
```